

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000830	Version:	4.0, CURRENT
	Status:	Effective	Effective Date:	14-Jul-2011
Title:	SM DG57-DG517. Electrovalves replacement.			Page 2 of 8

DG-57 / DG-517 Service Manual

ELECTROVALVES REPLACEMENT

Procedure description
This procedure describes how replace the electrovalves of the instrument. <i>Note: Document REGD-0000830_v4 replaces document REGD-0000830_v3.</i>

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EQUIPMENT

1. An instrument turned off.
 2. -231802 ELECTROVALVE 2 WAYS.
-231803 ELECTROVALVE 3 WAYS.
-or 232263 PCB ELECTROVALVES WITH VALVES (Entire electrovalve module).
 3. Welding maching with tin.

SET UP

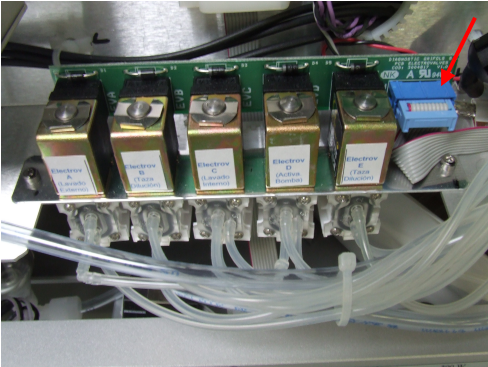
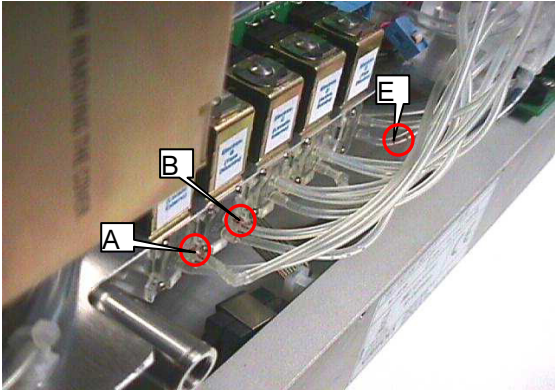
1. Remove the rear panel of the instrument.



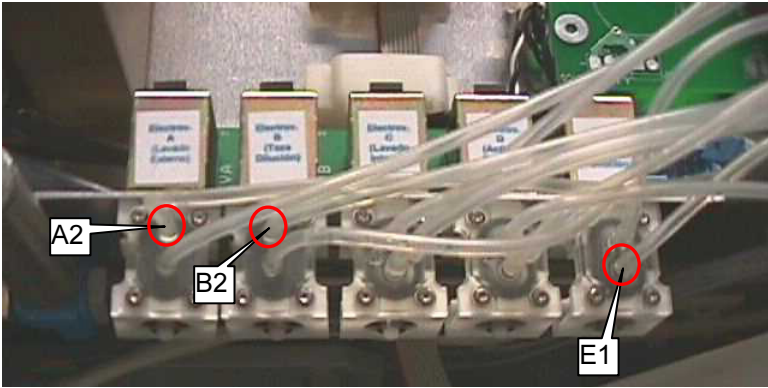
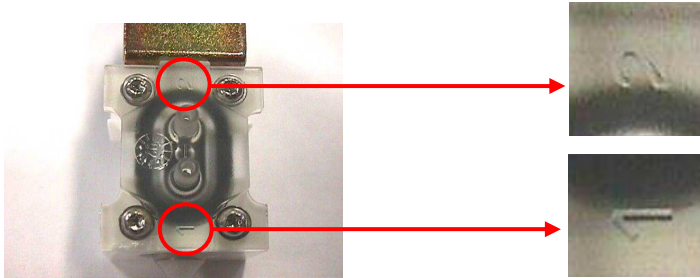
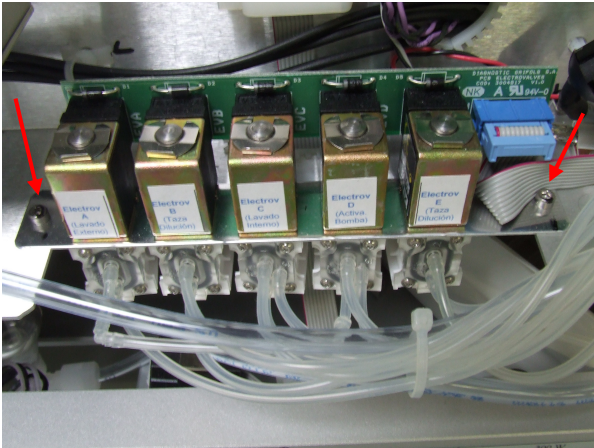
WARNING!

There may be rests of blood in the electrovalves. Perform decontamination procedure REGD-0000787 before changing the electrovalves.

PROCEDURE

Process	Expected Results
1. Disconnect the flat cable connector.  Picture 1	---
2. Mark with a permanent pen the tubes showed in picture 2 and 3. The text of the mark may be the ones showed in pictures 2 and 3.  Picture 2	---

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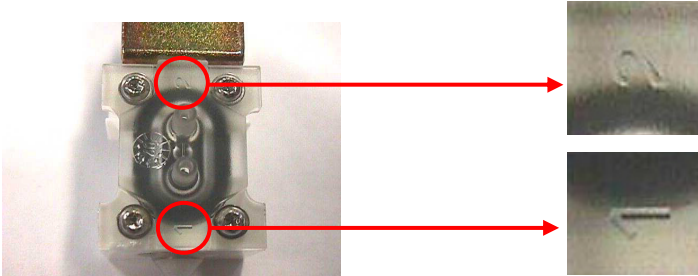
Process	Expected Results
 Picture 3 These texts correspond with the position of the electrovalve and the number of the connections in each electrovalve.  Picture 4 Picture 5 3. Unscrew the electrovalves module from the instrument by unscrewing two screws.  Picture 6 4. Open all the bottles of the instrument (Sol. A, Sol B and Waste). Extract all the tubes of the electrovalves.	---

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Process

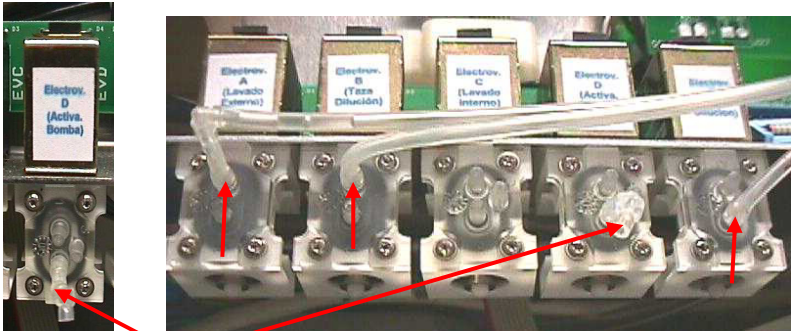
Expected Results

12. Each electrovalve has a correlation name, A, B, C, D and E.
Each electrovalve has a number in each connection.



Picture 10

Connect the tubes marked previously in the step 2 to the corresponding connections. (A2, B2 and E1).
Connect the cap tube in the electrovalve D position 2.



Picture 11

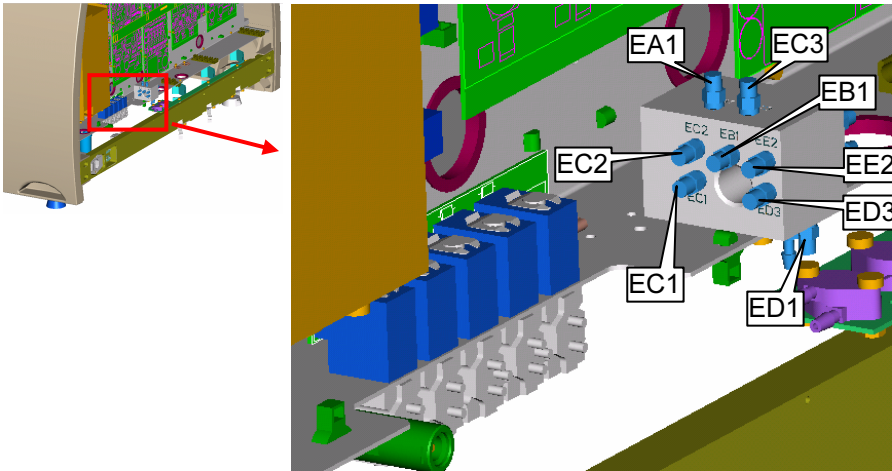
Connect the rest of the tubes following the text of the fluidic distributor.
for example: EA1 means:

EA 1

Electrovalve A

Connection 1

Picture 12



Picture 13

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Process	Expected Results
13. Connect the flat cable of the PCB (see Picture 1). 14. Verify that all the tubes are connected properly in its connection. Picture 14	--- ---
15. Verify that all the electrovalves connections have the appropriate tube connected (See picture 13). Picture 15	---
16. Close the bottles cap. Don't forget to replace the top and rear cover of the instrument.	---
17. Apply the REGD-0000821 procedure to check the fluid module.	---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	<i>Document REGD-0000830_v2 replaces document 57057V0M000.</i>	Document reference changed
3.0	2011/05/02	Added `previous version `data`, `change` and `rationale` inside document change history section.	Update change history for previous document history modification.
3.0	2011/05/02	Updated the Document Change History: Date format changed to YYYY/MM/DD Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
3.0	2011/05/02	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
3.0	2011/05/02	<i>Document REGD-0000830_v3 replaces document REGD-0000830_v2.</i>	Document version changed.
3.0	2011/05/02	Added new spare part code inside `equipment` sections.	Document change to update part numbers.
3.0	2011/05/02	Procedure modified.	Electrovalves are no longer installed directly to the rear part of the instrument. Electrovalve module has been created to easy electrovalve replacement.
4.0	See document header.	Document REGD-0000830_v4 replaces document REGD-0000830_v3.	Document version changed.
4.0	See document header.	Document title.	Commercial reference substituted by project reference.
4.0	See document header.	Document header.	Increase information to easy document search.